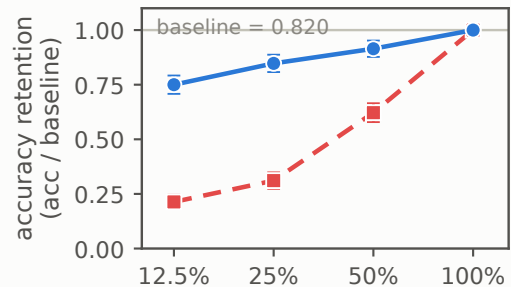


Retention vs compression depth

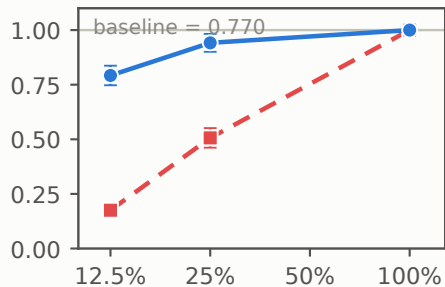
TextVQA

scene text



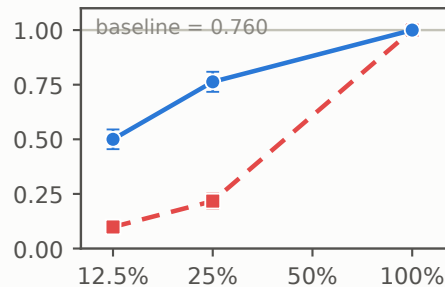
DocVQA

document



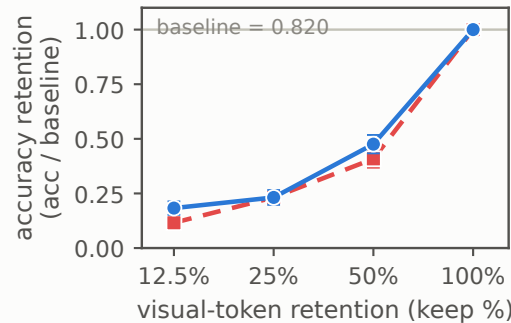
OCR-Bench

mixed OCR



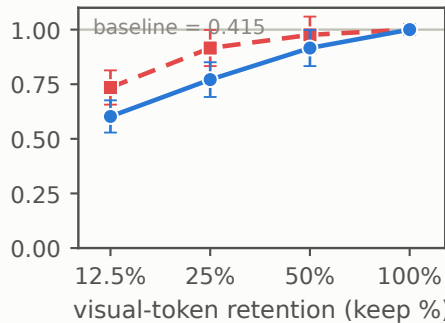
ChartQA

chart (budget-bound)



GQA

object



selection stage

pre-merger

post-merger

By tier:

- Text-dense (TextVQA, DocVQA, OCR-Bench): pre-merger keeps ~70-90% of baseline at 25% retention; post-merger collapses.
- ChartQA (budget-bound): both stages collapse — accuracy is token-budget limited, not selector-stage limited.
- GQA (object): post-merger is marginally stronger — the only tier where merging helps.